

Mini-Project Proposal

1. Project Title

(Provide a clear, specific title)

OutfitMatch: Multimodal Recommender

2. Team Members

(List all team members and briefly indicate roles or responsibilities (if assigned).)

3. Problem Statement and Motivation

Problem

Users need fast, personalized outfit suggestions that match free-text style requests (e.g., "casual summer, avoid black, budget <\$100") and optional uploaded photos. Existing solutions often rely on single-modality signals or simple similarity metrics and therefore fail to capture inter-item compatibility (how well pieces work together), user constraints (size, budget, exclusions), and explainability (why a suggestion is good or bad).

Motivation

A robust multimodal system that understands both text and images and models set-level compatibility can:

- Produce coherent, style-consistent outfit proposals rather than isolated similar items.
- Respect user constraints and personal context (uploaded photo, size, budget).
- Improve user satisfaction and conversion in commerce scenarios and advance research on multimodal alignment and set recommendation.

Deliverables

- Input: free-text style query + optional user photo + constraints (size, budget, exclusions).
- Output: Top-K outfit proposals (images + item IDs), compatibility scores, short textual rationale, and replacement suggestions (basic or advanced depending on scope).

4. Proposed Methodology

(Explain the type of AI task, and the models or algorithms you plan to use/develop)

Overview

This project is a multimodal retrieval and set-level scoring system that combines single-item extraction, image-text alignment, candidate retrieval, and outfit compatibility scoring.

Pipeline (high level)

- Item extraction: detect and segment garments from photos using Mask R-CNN; produce standardized item crops and basic attributes.
- Image and text encoding: encode item images with a visual backbone (ResNet/ViT) and user queries/product titles with a text encoder (DistilBERT/BERT); project both to a shared 512-d space.
- Cross-modal alignment: fine-tune with CLIP-style contrastive learning (InfoNCE) to align image and text embeddings for semantic retrieval.
- Retrieval: index item embeddings with FAISS and retrieve top-N candidates per category conditioned on text and hard constraints (size, budget, exclusions).

- Compatibility scoring: score candidate sets with a set-level model (recommended: GNN or Set Transformer) trained on outfit-level positives and generated negatives; output a compatibility score for ranking.
- Explanation & suggestions (advanced): attribute prediction + attribution (LOO/Shapley) to identify problematic items, rule-based conflict detection, and replacement retrieval with counterfactual re-scoring; generate template-based textual suggestions.

Training strategy

- Stage 1: Train detection/segmentation on DeepFashion2; extract item crops and attributes.
- Stage 2: Fine-tune visual encoder on product images and attribute tasks.
- Stage 3: Build image-text pairs and fine-tune CLIP-style alignment.
- Stage 4: Prepare outfit positives (e.g., Polyvore) and generate negatives (random, attribute-conflict, hard negatives); train compatibility model with BCE or ranking loss.
- Stage 5 (optional): integrate attribution and replacement retrieval; fine-tune end-to-end if resources permit.

Evaluation

- Retrieval: Recall@K, mAP.
- Compatibility: AUC, accuracy on held-out positive/negative outfit pairs.
- Human evaluation: subjective ratings for coherence and usefulness of suggestions.